Commutative Algebra 3

Splitting fields and residue fields of polynomials

Let k be a field, let $\overline{k}$ be a fixed algebraic closure, and let

$$g(t) \in k[t]$$

be a nonconstant polynomial. The purpose of this section is to explain precisely how the notion of a *splitting field* is related to the notion of a *residue field* of a point of $\mathrm{Spec}(k[t])$.

Main idea. A residue field of a closed point of $\mathrm{Spec}(k[t])$ usually gives the field generated by *one root*, whereas the splitting field is the smallest field containing *all roots*. Thus a splitting field is generally not the residue field of the original point (g), but it is built naturally from such residue fields, and in many cases it can itself be realized as a residue field of another point.

1. Residue fields in general

Let A be a commutative ring and let $\mathfrak{p} \in \mathrm{Spec}(A)$. The residue field at $\mathfrak{p}$ is

$$\kappa(\mathfrak{p}) := A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

If $\mathfrak{m}$ is maximal, then localization is unnecessary for the final field, and one has

$$\kappa(\mathfrak{m}) \cong A/\mathfrak{m}.$$

So for a closed point of an affine scheme, the residue field is simply the quotient by the corresponding maximal ideal.

2. Closed points of $\mathrm{Spec}(k[t])$

Now let

$$A = k[t].$$

Since $k[t]$ is a principal ideal domain, its prime ideals are exactly:

$$(0) \qquad \text{and} \qquad (f),$$

where $f \in k[t]$ is irreducible.

The maximal ideals are the ideals (f) with f irreducible and nonconstant. Therefore a closed point of

$$\mathbb{A}^1_k = \mathrm{Spec}(k[t])$$

is determined by an irreducible polynomial $f \in k[t]$, and its residue field is

$$\kappa((f)) \cong k[t]/(f).$$

If $\alpha \in \overline{k}$ is a root of f, then

$$k[t]/(f) \cong k(\alpha).$$

Hence the residue field of the point (f) is the field obtained by adjoining one root of f.

Important interpretation:

$$\boxed{\kappa((f)) \cong k(\alpha) \text{ is the field of definition of the corresponding root-point.}}$$

3. Algebraically closed base field

If k is algebraically closed, every irreducible polynomial in $k[t]$ is linear, so the maximal ideals are exactly

$$(t-a), \qquad a \in k.$$

Then

$$\kappa((t-a)) \cong k[t]/(t-a) \cong k.$$

Thus over an algebraically closed field all closed points of $\mathbb{A}^1_k$ are k-rational, and their residue fields are just k.

The generic point is (0), whose residue field is

$$\kappa((0)) \cong k(t),$$

the rational function field. This is transcendental over k, not algebraic.

4. Splitting field of a polynomial

Let

$$g(t) \in k[t]$$

have roots $\alpha_1, \dots, \alpha_n$ in $\overline{k}$, counted without regard to multiplicity. The *splitting field* of g over k is the smallest subfield

$$L \subseteq \overline{k}$$

such that all roots α_i lie in L. Equivalently,

$$L = k(\alpha_1, \dots, \alpha_n).$$

So the splitting field is obtained by adjoining *all* roots at once, not just one of them.

Key distinction:

$$\boxed{\text{Residue field of a closed point} = \text{field generated by one root}}$$

whereas

$$\boxed{\text{Splitting field} = \text{smallest field containing all roots.}}$$

5. Why residue field and splitting field are usually different

Suppose $f \in k[t]$ is irreducible and let α be one of its roots. Then

$$\kappa((f)) \cong k[t]/(f) \cong k(\alpha).$$

However, the other roots of f need not lie in $k(\alpha)$. Therefore in general

$$k(\alpha) \subsetneq \text{splitting field of } f.$$

So the residue field of the point (f) usually captures only one algebraic branch of the root structure, while the splitting field captures the whole family of conjugate roots.

6. First example: $x^2 - 2$ over $\mathbb{Q}$

Take

$$f(x) = x^2 - 2 \in \mathbb{Q}[x].$$

This polynomial is irreducible over $\mathbb{Q}$, so the point $(x^2 - 2)$ is a closed point of $\operatorname{Spec}(\mathbb{Q}[x])$, and

$$\kappa((x^2 - 2)) \cong \mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}).$$

The roots of $x^2 - 2$ are

$$\sqrt{2} \qquad \text{and} \qquad -\sqrt{2},$$

and both already lie in $\mathbb{Q}(\sqrt{2})$. Therefore the residue field and the splitting field coincide:

$$\kappa((x^2 - 2)) = \mathbb{Q}(\sqrt{2}) = \text{splitting field of } x^2 - 2.$$

So equality can happen.

7. Second example: $x^3 - 2$ over $\mathbb{Q}$

Now take

$$f(x) = x^3 - 2 \in \mathbb{Q}[x].$$

This polynomial is irreducible over $\mathbb{Q}$, so again

$$\kappa((x^3 - 2)) \cong \mathbb{Q}[x]/(x^3 - 2) \cong \mathbb{Q}(\sqrt[3]{2}).$$

But the three roots are

$$\sqrt[3]{2}, \qquad \omega\sqrt[3]{2}, \qquad \omega^2\sqrt[3]{2},$$

where ω is a primitive cube root of unity. Hence the splitting field is

$$\mathbb{Q}(\sqrt[3]{2}, \omega),$$

which is strictly larger than $\mathbb{Q}(\sqrt[3]{2})$.

Thus in this example

$$\kappa((x^3 - 2)) = \mathbb{Q}(\sqrt[3]{2}) \neq \mathbb{Q}(\sqrt[3]{2}, \omega) = \text{splitting field}.$$

This shows very clearly that the residue field of the point (f) need not be the splitting field.

8. Residue fields of irreducible factors

Suppose
$$g(t) = f_1(t) \cdots f_r(t)$$
is the factorization of g into distinct irreducible factors over k. For each i, the ideal (f_i) defines a closed point of $\mathrm{Spec}(k[t])$, and its residue field is

$$\kappa((f_i)) \cong k[t]/(f_i) \cong k(\alpha_i),$$

where α_i is a chosen root of f_i.

Then the splitting field L of g is the smallest field containing all these root fields:

$$L = k(\alpha_1, \ldots, \alpha_r).$$

So one may say:

$$\boxed{\text{The splitting field is the compositum of the residue fields attached to the irreducible factors.}}$$

This is the most direct and useful way to understand a splitting field in terms of residue fields.

9. Geometric meaning on the affine line

Consider the zero-locus of g:

$$V(g) \subseteq \mathbb{A}^1_k = \mathrm{Spec}(k[t]).$$

Over the base field k, this closed subscheme may fail to decompose into k-rational points. Its closed points can instead have residue fields strictly larger than k.

For example, over $\mathbb{R}$,
$$t^2 + 1$$
has no real root, so the corresponding point is not an $\mathbb{R}$-rational point. Instead its residue field is
$$\mathbb{R}[t]/(t^2 + 1) \cong \mathbb{C}.$$

Thus the residue field of a closed point records the smallest field over which that point is visible as an actual rational root.

The splitting field of g is then the smallest extension field over which *all* roots become rational points simultaneously.

So geometrically:

$$\boxed{\text{The splitting field is the smallest field over which the zero-scheme } V(g) \text{ splits into visible root-points.}}$$

10. Base change to the splitting field

Let L be the splitting field of g. Then in $L[t]$ one has

$$g(t) = c \prod_{j=1}^{n} (t - \alpha_j).$$

Passing from k to L means performing base change:

$$V(g)_L := V(g) \times_{\mathrm{Spec}(k)} \mathrm{Spec}(L).$$

On coordinate rings this corresponds to

$$k[t]/(g) \otimes_k L \cong L[t]/(g).$$

Now, if g has no repeated roots, then

$$L[t]/(g) \cong \prod_{j=1}^{n} L$$

by the Chinese remainder theorem, because

$$(g) = \prod_{j=1}^{n} (t - \alpha_j)$$

with pairwise comaximal linear factors.

This means that after extension to the splitting field, the quotient algebra decomposes into one copy of L for each root.

So the splitting field is exactly the field over which the algebra becomes completely decomposed into pieces corresponding to individual roots.

11. Example: $t^2 + 1$ over $\mathbb{R}$

Let

$$g(t) = t^2 + 1 \in \mathbb{R}[t].$$

Since $t^2 + 1$ is irreducible over $\mathbb{R}$, the corresponding closed point has residue field

$$\kappa((t^2 + 1)) \cong \mathbb{R}[t]/(t^2 + 1) \cong \mathbb{C}.$$

Its splitting field is also $\mathbb{C}$, because

$$t^2 + 1 = (t - i)(t + i)$$

in $\mathbb{C}[t]$.

After base change,

$$\mathbb{R}[t]/(t^2 + 1) \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[t]/(t^2 + 1) \cong \mathbb{C}[t]/((t - i)(t + i)) \cong \mathbb{C} \times \mathbb{C}.$$

Thus over $\mathbb{R}$ we have one closed point with residue field $\mathbb{C}$, while over $\mathbb{C}$ that point splits into two actual $\mathbb{C}$-points.

12. Splitting field as a residue field via a primitive element

Although the splitting field is usually not the residue field of the original point $(g) \subseteq k[t]$, it can often be realized as a residue field of another point.

Suppose L/k is a finite separable extension. Then by the primitive element theorem there exists $\theta \in L$ such that

$$L = k(\theta).$$

Let $m_\theta(u) \in k[u]$ be the minimal polynomial of θ. Then

$$L \cong k[u]/(m_\theta(u)).$$

Hence L is the residue field of the closed point $(m_\theta) \in \mathrm{Spec}(k[u])$.
So for finite separable splitting fields one has:

$$\boxed{\text{The splitting field can itself be realized as a residue field of a closed point of an affine line.}}$$

However, this point is associated with a primitive element θ, not necessarily with the original polynomial g.

13. Splitting field as the residue field of a point encoding all roots

There is another, more geometric, way to realize the splitting field as a residue field.
Suppose g has degree n, and let its roots in $\overline{k}$ be

$$\alpha_1, \dots, \alpha_n.$$

Introduce variables $x_1, \dots, x_n$, intended to represent the roots, and consider an affine scheme whose points encode tuples of roots satisfying

$$\prod_{j=1}^{n} (t - x_j) = g(t).$$

Equating coefficients gives polynomial equations in the x_j, so one obtains a finitely generated k-algebra whose points correspond to ordered root-tuples. The point

$$(\alpha_1, \dots, \alpha_n)$$

has residue field

$$k(\alpha_1, \dots, \alpha_n),$$

which is precisely the splitting field L.
So in a larger affine space, one may say:

$$\boxed{\text{The splitting field is the residue field of the point representing all roots simultaneously.}}$$

This gives an exact geometric realization of the splitting field as a residue field.

14. Relation with the generic point

It is useful to contrast all of this with the generic point of $\mathrm{Spec}(k[t])$, namely the prime ideal (0). Its residue field is

$$\kappa((0)) \cong k(t),$$

the rational function field.

This field is not obtained by adjoining algebraic roots of a fixed polynomial. It corresponds instead to the generic point of the whole affine line. Therefore residue fields of $\mathrm{Spec}(k[t])$ come in two very different kinds:

- for closed points (f), one gets algebraic extensions

$$\kappa((f)) \cong k[t]/(f) \cong k(\alpha);$$

- for the generic point (0), one gets the transcendental extension

$$\kappa((0)) \cong k(t).$$

So the notion of residue field is much broader than that of splitting field.

15. Example with several irreducible factors

Consider

$$g(t) = t^4 - 5t^2 + 6 = (t^2 - 2)(t^2 - 3) \in \mathbb{Q}[t].$$

The irreducible factors give two closed points:

$$(t^2 - 2), \qquad (t^2 - 3).$$

Their residue fields are

$$\kappa((t^2 - 2)) \cong \mathbb{Q}(\sqrt{2}), \qquad \kappa((t^2 - 3)) \cong \mathbb{Q}(\sqrt{3}).$$

The splitting field of g is

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

So here the splitting field is visibly the compositum of the residue fields attached to the irreducible factors. This is an especially transparent example of the general principle.

16. Galois-theoretic interpretation

Let $f \in k[t]$ be irreducible, let α be one root, and let

$$k(\alpha) \cong k[t]/(f)$$

be the corresponding residue field. The other roots of f are the conjugates of α under embeddings into $\overline{k}$. The splitting field is the smallest field containing all these conjugates.

Thus:

- the residue field of (f) is generated by one chosen conjugate;
- the splitting field is generated by the whole orbit of conjugates.

From the Galois point of view, the residue field captures one branch, while the splitting field captures the full symmetry of the polynomial.

17. Final summary

Let $g(t) \in k[t]$.

Pointwise picture. For each irreducible factor f of g, the closed point $(f) \subseteq \operatorname{Spec}(k[t])$ has residue field

$$\kappa((f)) \cong k[t]/(f) \cong k(\alpha),$$

where α is one root of f.

Global picture. The splitting field of g is the smallest field containing all such root fields:

$$L = k(\alpha_1, \dots, \alpha_r).$$

Geometric picture. The splitting field is the smallest extension over which the zero-scheme $V(g)$ decomposes into actual rational root-points.

Residue-field realization. Although L is usually not the residue field of the original point (g), it can often be realized as a residue field:

- of a closed point corresponding to a primitive element of L/k, or
- of a point in a larger affine space encoding all roots simultaneously.

Therefore the clean conceptual statement is:

$$\boxed{\text{Residue field of a closed point} = \text{field of definition of one root-point,}}$$

while

$$\boxed{\text{Splitting field} = \text{smallest field over which all root-points are simultaneously defined.}}$$

18. Scheme-theoretic interpretation of "one point" versus "all geometric points"

Let

$$X = V(g) \subseteq \mathbb{A}^1_k.$$

Over the base field k, a closed point of X can have residue field strictly larger than k. Such a point is then not necessarily a k-rational point.

Over the algebraic closure $\overline{k}$, every root of g becomes an actual $\overline{k}$-point:

$$X(\overline{k}) = \{\alpha_1, \dots, \alpha_n\}.$$

What the splitting field does is to find the smallest field

$$L, \qquad k \subseteq L \subseteq \overline{k},$$

such that all these geometric points are already L-rational.

So one gets the following very clean geometric interpretation:

- a residue field tells you where a single closed point lives;
- the splitting field tells you where all geometric points become simultaneously rational.

Equivalently, if one thinks of the roots of g as geometric points of the zero-scheme $V(g)$, then the splitting field is the smallest extension of the base field over which all those geometric points are simultaneously visible as ordinary rational points.

19. A precise slogan

The cleanest slogan is:

$$\boxed{\text{The residue field of a closed point of } \operatorname{Spec}(k[t]) \text{ is the field generated by one root.}}$$

and

$$\boxed{\text{The splitting field of a polynomial is the smallest field containing the residue fields of all root-points.}}$$

Or, stated even more conceptually:

$$\boxed{\text{Residue field} = \text{field of definition of one point,} \qquad \text{splitting field} = \text{field of definition of all roots at on}}$$

These formulas capture the exact difference between the local pointwise information carried by a residue field and the global root information carried by the splitting field.

20. Final conceptual summary

Let

$$g(t) \in k[t].$$

First layer: pointwise.
For each irreducible factor f of g, the closed point

$$(f) \subseteq \operatorname{Spec}(k[t])$$

has residue field

$$\kappa((f)) \cong k[t]/(f) \cong k(\alpha),$$

where α is one root of f.

Second layer: global root field.
The splitting field is

$$L = k(\alpha_1, \dots, \alpha_r),$$

the smallest field containing all these root fields.

Third layer: after base change.
Over L, the polynomial splits completely, and the zero-scheme decomposes into actual L-rational root-points.

Fourth layer: realization as a residue field.
Although L is usually not the residue field of the original point (g), it can often be realized as a residue field:

- of a point corresponding to a primitive element in some affine line;
- or of a point in a larger affine space that records all roots simultaneously.

So the conceptual hierarchy is:

$$\text{one closed point} \longrightarrow \text{one residue field} \longrightarrow \text{one root field},$$

whereas

$$\text{all geometric root-points} \longrightarrow \text{splitting field}.$$

21. One compact answer to the original question

If someone asks:

> Can we explain the splitting field of a polynomial as a residue field?

the best precise answer is:

Yes, but with care.
For an irreducible polynomial f, the residue field

$$k[t]/(f)$$

is the field obtained by adjoining one root.

The splitting field is generally larger: it is the smallest field containing all conjugate roots, hence the compositum of the relevant residue fields.

In many cases it can itself be realized as a residue field of another closed point, for example:

- via a primitive element;
- or via a point encoding all roots simultaneously.

So the most accurate compact statement is:

$$\boxed{\text{A residue field usually gives one root, while the splitting field gives all roots together.}}$$

And therefore:

$$\boxed{\text{The splitting field is not usually the residue field of the original point,}}$$

but

$$\boxed{\text{it is naturally built from residue fields and can often be realized as a residue field in a larger or differe}}$$

Problem 6 — Existence of the inverse limit

Exact statement.

We start with a special case of inverse limits (which are themselves a special case of the notion of limit in category theory). Suppose given a collection G_n, $n \geq 1$, of abelian groups equipped with homomorphisms

$$\phi_n : G_{n+1} \to G_n \qquad (n \geq 1).$$

We call such data an inverse system.

Then the inverse limit of $\{G_n\}$ is an abelian group G equipped with maps

$$g_n : G \to G_n$$

for each n, with

$$\phi_n \circ g_{n+1} = g_n,$$

satisfying the following universal property:

Given any abelian group H and homomorphisms

$$h_n : H \to G_n$$

with

$$\phi_n \circ h_{n+1} = h_n,$$

there exists a unique homomorphism

$$h : H \to G$$

with

$$h_n = g_n \circ h$$

for all n.

Show that the inverse limit exists. Show further that if each G_n has the structure of a ring (resp. A-module) and the ϕ_n are ring homomorphisms (resp. A-module homomorphisms), then the inverse limit is also a ring or A-module.

We write

$$\varprojlim G_n$$

for the inverse limit.

Solution.

Define

$$G := \left\{ (x_1, x_2, \dots) \in \prod_{n\geq 1} G_n \;\middle|\; \phi_n(x_{n+1}) = x_n \text{ for all } n \geq 1 \right\}.$$

This is a subgroup of the direct product $\prod_{n\geq 1} G_n$, since the compatibility condition is preserved under addition and inverses.

For each n, define

$$g_n : G \to G_n, \qquad g_n((x_m)_{m\geq 1}) = x_n.$$

Then, by definition of G,

$$\phi_n \circ g_{n+1}(x) = \phi_n(x_{n+1}) = x_n = g_n(x),$$

so the maps g_n are compatible.

We now verify the universal property. Let H be an abelian group and suppose we are given homomorphisms

$$h_n : H \to G_n$$

such that

$$\phi_n \circ h_{n+1} = h_n \qquad \text{for all } n.$$

Define

$$h : H \to G, \qquad h(y) := (h_1(y), h_2(y), h_3(y), \dots).$$

This is well-defined, because

$$\phi_n(h_{n+1}(y)) = h_n(y)$$

for every n, so $(h_n(y))_{n\geq 1}$ is a compatible sequence. Clearly h is a homomorphism, and

$$g_n \circ h = h_n$$

for every n.

To prove uniqueness, suppose $h' : H \to G$ is another homomorphism with

$$g_n \circ h' = h_n \qquad \text{for all } n.$$

Then for each $y \in H$, all coordinates of $h'(y)$ are determined:

$$g_n(h'(y)) = h_n(y).$$

Hence

$$h'(y) = (h_1(y), h_2(y), \dots) = h(y),$$

so $h' = h$. Therefore G is the inverse limit.

Now assume that each G_n is a ring and each ϕ_n is a ring homomorphism. Then $\prod_{n\geq 1} G_n$ is a ring with componentwise operations. If $x = (x_n)$ and $y = (y_n)$ lie in G, then

$$\phi_n(x_{n+1}y_{n+1}) = \phi_n(x_{n+1})\phi_n(y_{n+1}) = x_n y_n,$$

so $xy \in G$. Thus G is a subring of $\prod G_n$. Its identity is $(1, 1, 1, \dots)$.

Similarly, if each G_n is an A-module and each ϕ_n is A-linear, then $\prod_{n\geq 1} G_n$ is an A-module componentwise, and for $a \in A$,

$$\phi_n(ax_{n+1}) = a\,\phi_n(x_{n+1}) = ax_n,$$

so G is an A-submodule.

Therefore the inverse limit exists, and in the ring or A-module setting it inherits the corresponding structure.

Problem 7 — Exactness of inverse limits

Exact statement.

Let $\{A_n\}$, $\{B_n\}$, $\{C_n\}$ be three inverse systems of abelian groups as in Question 6, forming an exact sequence in the sense that for each n there is a commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & A_{n+1} & \to & B_{n+1} & \to & C_{n+1} & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & A_n & \to & B_n & \to & C_n & \to & 0 \end{array}$$

where the vertical maps are the maps of the three inverse systems and the rows are short exact sequences.

Show there is an exact sequence

$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n.$$

Show furthermore that the right-most map is surjective if the inverse system $\{A_n\}$ has all maps

$$\phi_n : A_{n+1} \to A_n$$

surjective.

Solution.
Let

$$i_n : A_n \to B_n, \qquad p_n : B_n \to C_n$$

be the maps in the short exact rows, and let the transition maps be

$$\alpha_n : A_{n+1} \to A_n, \qquad \beta_n : B_{n+1} \to B_n, \qquad \gamma_n : C_{n+1} \to C_n.$$

These induce homomorphisms

$$i : \varprojlim A_n \to \varprojlim B_n, \qquad i((a_n)) = (i_n(a_n)),$$

and

$$p : \varprojlim B_n \to \varprojlim C_n, \qquad p((b_n)) = (p_n(b_n)).$$

We prove that

$$0 \to \varprojlim A_n \xrightarrow{i} \varprojlim B_n \xrightarrow{p} \varprojlim C_n$$

is exact.

First, i is injective. Indeed, if $i((a_n)) = 0$, then

$$i_n(a_n) = 0 \qquad \text{for all } n.$$

Since each i_n is injective, it follows that $a_n = 0$ for all n, hence $(a_n) = 0$.

Next, $\mathrm{Im}(i) \subseteq \ker(p)$, since for $(a_n) \in \varprojlim A_n$,

$$p(i((a_n))) = (p_n(i_n(a_n)))_n = (0)_n$$

because $p_n \circ i_n = 0$ in every short exact row.

Now let $(b_n) \in \varprojlim B_n$ lie in $\ker(p)$. Then

$$p_n(b_n) = 0 \qquad \text{for all } n.$$

By exactness of each row, there exists a unique $a_n \in A_n$ such that

$$i_n(a_n) = b_n.$$

We claim that (a_n) is compatible. Indeed,

$$i_n(\alpha_n(a_{n+1})) = \beta_n(i_{n+1}(a_{n+1})) = \beta_n(b_{n+1}) = b_n = i_n(a_n).$$

Since i_n is injective,

$$\alpha_n(a_{n+1}) = a_n.$$

Thus $(a_n) \in \varprojlim A_n$, and clearly $i((a_n)) = (b_n)$. Therefore

$$\ker(p) = \mathrm{Im}(i).$$

So inverse limit is left exact.

Assume now that every transition map

$$\alpha_n : A_{n+1} \to A_n$$

is surjective. We show that
$$p : \varprojlim B_n \to \varprojlim C_n$$
is surjective.

Take $(c_n) \in \varprojlim C_n$. We construct a compatible sequence (b_n) with
$$p_n(b_n) = c_n \qquad \text{for all } n.$$

Choose $b_1 \in B_1$ such that
$$p_1(b_1) = c_1.$$
Suppose $b_n \in B_n$ has been chosen with $p_n(b_n) = c_n$. Choose any
$$b'_{n+1} \in B_{n+1}$$
such that
$$p_{n+1}(b'_{n+1}) = c_{n+1}.$$
Then
$$p_n(\beta_n(b'_{n+1}) - b_n) = \gamma_n(c_{n+1}) - c_n = 0,$$
since (c_n) is compatible. By exactness of
$$0 \to A_n \xrightarrow{i_n} B_n \xrightarrow{p_n} C_n \to 0,$$
there exists $a_n \in A_n$ such that
$$i_n(a_n) = \beta_n(b'_{n+1}) - b_n.$$
Because α_n is surjective, choose $a_{n+1} \in A_{n+1}$ with
$$\alpha_n(a_{n+1}) = a_n.$$
Set
$$b_{n+1} := b'_{n+1} - i_{n+1}(a_{n+1}).$$
Then
$$p_{n+1}(b_{n+1}) = p_{n+1}(b'_{n+1}) = c_{n+1},$$
and
$$\beta_n(b_{n+1}) = \beta_n(b'_{n+1}) - \beta_n(i_{n+1}(a_{n+1})) = \beta_n(b'_{n+1}) - i_n(\alpha_n(a_{n+1})) = \beta_n(b'_{n+1}) - i_n(a_n) = b_n.$$
Thus (b_n) is compatible, so $(b_n) \in \varprojlim B_n$, and
$$p((b_n)) = (c_n).$$
Hence p is surjective.

Therefore, under the additional hypothesis that all maps $A_{n+1} \to A_n$ are surjective, we obtain a short exact sequence
$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0.$$

Problem 8 — Completion as an inverse limit

Exact statement.

A frequent situation is that an inverse system arises from a filtration, i.e.

$$G \supset G_1 \supset G_2 \supset \cdots,$$

so that there are natural surjections

$$\phi_n : G/G_{n+1} \to G/G_n,$$

hence giving

$$\varprojlim G/G_n.$$

Here we give a topological interpretation. First, we may put a topology on G with basis of open sets given by

$$\{\, g + G_n \mid g \in G,\ n \geq 1\,\}.$$

Show that this topology is Hausdorff if

$$\bigcap_{n=1}^{\infty} G_n = 0.$$

We may define the completion of G with respect to this topology in the usual way. A sequence g_n in G is *Cauchy* if for any open neighbourhood $U \subseteq G$ of 0, there exists an N such that for all $n, n' \geq N$,

$$g_n - g_{n'} \in U.$$

Two Cauchy sequences g_n, g'_n are *equivalent* if for each open neighbourhood U of 0, there exists an N such that for all $n \geq N$,

$$g_n - g'_n \in U.$$

The completion $\widehat{G}$ of G is then defined to be the set of Cauchy sequences of G modulo equivalence.

Show that $\widehat{G}$ has the structure of an abelian group, and there is a canonical map

$$G \to \widehat{G}$$

with kernel

$$\bigcap_{n=1}^{\infty} G_n.$$

Finally, show there is an isomorphism of abelian groups

$$\varprojlim G/G_n \cong \widehat{G},$$

thus showing the inverse limit does not depend on the filtration but only on the induced topology.

Solution.

We proceed in several steps.

1. The topology is Hausdorff.
The basic open neighbourhoods of 0 are precisely the subgroups G_n. Assume

$$\bigcap_{n=1}^{\infty} G_n = 0.$$

Let $x, y \in G$ with $x \neq y$. Then $x - y \neq 0$, so there exists n such that

$$x - y \notin G_n.$$

We claim that the basic open sets $x + G_n$ and $y + G_n$ are disjoint. Indeed, if they had a common point, then for some $u, v \in G_n$,

$$x + u = y + v,$$

hence

$$x - y = v - u \in G_n,$$

a contradiction. Therefore distinct points can be separated by disjoint open neighbourhoods, so the topology is Hausdorff.

2. The completion $\widehat{G}$ is an abelian group.
Let (x_m) and (y_m) be Cauchy sequences in G. Define

$$(x_m) + (y_m) := (x_m + y_m).$$

We show that $(x_m + y_m)$ is again Cauchy. Let G_r be a basic neighbourhood of 0. Since (x_m) is Cauchy, there exists N_1 such that

$$x_m - x_{m'} \in G_r \qquad (m, m' \geq N_1).$$

Similarly, there exists N_2 such that

$$y_m - y_{m'} \in G_r \qquad (m, m' \geq N_2).$$

Hence for $m, m' \geq \max(N_1, N_2)$,

$$(x_m + y_m) - (x_{m'} + y_{m'}) = (x_m - x_{m'}) + (y_m - y_{m'}) \in G_r.$$

So $(x_m + y_m)$ is Cauchy.

Also $(-x_m)$ is Cauchy, so Cauchy sequences form an abelian group under termwise addition.

Now let $(x_m) \sim (x'_m)$ and $(y_m) \sim (y'_m)$. For every r, eventually

$$x_m - x'_m \in G_r \qquad \text{and} \qquad y_m - y'_m \in G_r.$$

Therefore eventually

$$(x_m + y_m) - (x'_m + y'_m) \in G_r,$$

so $(x_m + y_m) \sim (x'_m + y'_m)$. Thus addition descends to equivalence classes. The same holds for inverses. Hence $\widehat{G}$ is an abelian group.

3. The canonical map $G \to \widehat{G}$.

Define

$$\iota : G \to \widehat{G}$$

by sending $g \in G$ to the class of the constant sequence

$$(g, g, g, \dots).$$

This is a homomorphism.

Its kernel consists of those $g \in G$ such that the constant sequence $(g, g, \dots)$ is equivalent to the zero sequence $(0, 0, \dots)$. By definition, this means that for every neighbourhood U of 0, eventually

$$g \in U.$$

Since the sequence is constant, this is equivalent to saying $g \in U$ for every neighbourhood U of 0. In particular,

$$g \in G_n \qquad \text{for all } n.$$

Thus

$$\ker(\iota) = \bigcap_{n=1}^{\infty} G_n.$$

4. Construction of the map $\widehat{G} \to \varprojlim G/G_n$.

Let (x_m) be a Cauchy sequence in G. Fix n. Since (x_m) is Cauchy, there exists N such that for all $m, m' \geq N$,

$$x_m - x_{m'} \in G_n.$$

Hence the cosets $x_m + G_n$ are eventually constant. Let their eventual value be $a_n \in G/G_n$.

This produces a sequence (a_n) with

$$a_n \in G/G_n.$$

Moreover the compatibility condition holds:

$$\phi_n(a_{n+1}) = a_n,$$

because passing from G/G_{n+1} to G/G_n just reduces modulo G_n. Therefore

$$(a_n) \in \varprojlim G/G_n.$$

Equivalent Cauchy sequences define the same compatible system, so we obtain a well-defined homomorphism

$$\Psi : \widehat{G} \to \varprojlim G/G_n.$$

5. Construction of the map $\varprojlim G/G_n \to \widehat{G}$.

Take

$$(a_n) \in \varprojlim G/G_n.$$

Choose representatives $x_n \in G$ such that

$$a_n = x_n + G_n.$$

Since $\phi_n(a_{n+1}) = a_n$, the images of x_{n+1} and x_n in G/G_n agree, so

$$x_{n+1} - x_n \in G_n.$$

We claim that (x_n) is Cauchy. Fix r. If $m > n \geq r$, then

$$x_m - x_n = (x_m - x_{m-1}) + \cdots + (x_{n+1} - x_n).$$

For each $k \geq r$, we have

$$x_{k+1} - x_k \in G_k \subseteq G_r,$$

because the filtration is descending. Hence

$$x_m - x_n \in G_r.$$

So (x_n) is Cauchy.

If x'_n is another representative of a_n, then

$$x'_n - x_n \in G_n.$$

Therefore for each fixed r, whenever $n \geq r$,

$$x'_n - x_n \in G_n \subseteq G_r.$$

So $(x'_n) \sim (x_n)$. Thus the class of (x_n) in $\widehat{G}$ is independent of the choice of representatives.

Hence we get a well-defined homomorphism

$$\Phi : \varprojlim G/G_n \to \widehat{G}.$$

6. The two maps are inverse.

Let $[x_m] \in \widehat{G}$. Under Ψ, it maps to the compatible system whose n-th component is the eventual value of $x_m + G_n$. Reconstructing a Cauchy sequence from these eventual values gives a sequence equivalent to (x_m). Hence

$$\Phi \circ \Psi = \mathrm{id}_{\widehat{G}}.$$

Conversely, let $(a_n) \in \varprojlim G/G_n$, choose representatives x_n with

$$x_{n+1} - x_n \in G_n,$$

and form the Cauchy class $[x_n]$. For fixed r, if $n \geq r$, then

$$x_n - x_r \in G_r,$$

so

$$x_n + G_r = x_r + G_r.$$

Thus the eventual value modulo G_r is exactly a_r. Therefore

$$\Psi(\Phi((a_n))) = (a_n).$$

So

$$\Psi \circ \Phi = \mathrm{id}_{\varprojlim G/G_n}.$$

Therefore Φ and Ψ are inverse isomorphisms, and we conclude that

$$\varprojlim G/G_n \cong \widehat{G}.$$

This shows that the inverse limit depends only on the topology induced by the filtration, not on the filtration itself.

Problem 9 — I-adic completion of rings and modules

Exact statement.

Given a ring A and an ideal $I \subseteq A$, we obtain an inverse system with

$$A_n = A/I^n$$

and

$$\phi_n : A/I^{n+1} \to A/I^n$$

the obvious surjection. We write

$$\widehat{A} := \varprojlim A_n,$$

but bear in mind this depends on I. We call $\widehat{A}$ the I-adic completion of A.

Similarly, if M is an A-module, we obtain an inverse system of modules

$$M_n := M/I^n M,$$

and write $\widehat{M}$ for the inverse limit of the M_n, the I-adic completion of M.

(a) Describe the ring $k[[x_1, \ldots, x_n]]$, the completion of $k[x_1, \ldots, x_n]$ with respect to the maximal ideal $(x_1, \ldots, x_n)$. Describe the ring $\mathbf{Z}_p$, the completion of the integers $\mathbf{Z}$. [The former ring is called the ring of formal power series and the latter ring is called the ring of p-adic numbers.]

(b) Show that if A is Noetherian and

$$0 \to M_1 \to M_2 \to M_3 \to 0$$

is an exact sequence of finitely generated A-modules, then

$$0 \to \widehat{M_1} \to \widehat{M_2} \to \widehat{M_3} \to 0$$

is an exact sequence of $\widehat{A}$-modules.

(c) Show there is a canonical homomorphism

$$M \to \widehat{M},$$

and show the kernel of this homomorphism is

$$\bigcap_{n=0}^{\infty} I^n M.$$

(d) Show for M an A-module, there is a homomorphism

$$\widehat{A} \otimes_A M \to \widehat{M}$$

given as a composition

$$\widehat{A} \otimes_A M \to \widehat{A} \otimes_A \widehat{M} \to \widehat{A} \otimes_{\widehat{A}} \widehat{M} \cong \widehat{M}.$$

Show that if M is finitely generated, then this homomorphism is surjective. If in addition A is Noetherian, this homomorphism is an isomorphism.

Solution.

(a) Description of $k[[x_1, \ldots, x_n]]$ and $\mathbf{Z}_p$.

Let
$$A = k[x_1, \ldots, x_n], \qquad \mathfrak{m} = (x_1, \ldots, x_n).$$
Then
$$\widehat{A} = \varprojlim_r A/\mathfrak{m}^r.$$
The quotient $A/\mathfrak{m}^r$ records only terms of total degree $< r$. Hence an element of the inverse limit is exactly a compatible family of truncations of all orders, that is, a formal power series
$$\sum_{\alpha \in \mathbf{N}^n} a_\alpha x^\alpha, \qquad x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}.$$
Therefore
$$\widehat{k[x_1, \ldots, x_n]} \cong k[[x_1, \ldots, x_n]].$$

Now let
$$A = \mathbf{Z}, \qquad I = (p).$$
Then
$$\widehat{A} = \varprojlim_n \mathbf{Z}/p^n\mathbf{Z} =: \mathbf{Z}_p.$$
An element of $\mathbf{Z}_p$ is a compatible system
$$(a_n)_{n \geq 1}, \qquad a_n \in \mathbf{Z}/p^n\mathbf{Z},$$
such that $a_{n+1} \equiv a_n \pmod{p^n}$. Equivalently, each element of $\mathbf{Z}_p$ has a unique expansion
$$a_0 + a_1 p + a_2 p^2 + \cdots, \qquad a_i \in \{0, 1, \ldots, p-1\}.$$
Thus $\mathbf{Z}_p$ is the ring of p-adic integers.

(b) Exactness of completion for finitely generated modules over a Noetherian ring.

Assume
$$0 \to M_1 \xrightarrow{u} M_2 \xrightarrow{v} M_3 \to 0$$
is exact, A is Noetherian, and all modules are finitely generated.

For each n, since $M_3 \cong M_2/M_1$, we have
$$M_3/I^n M_3 \cong M_2/(M_1 + I^n M_2).$$
Hence there is a short exact sequence
$$0 \to \frac{M_1}{M_1 \cap I^n M_2} \to \frac{M_2}{I^n M_2} \to \frac{M_3}{I^n M_3} \to 0.$$

Passing to inverse limits and using the exactness result for inverse limits from the previous exercise, we obtain
$$0 \to \varprojlim \frac{M_1}{M_1 \cap I^n M_2} \to \widehat{M_2} \to \widehat{M_3} \to 0,$$

because the transition maps

$$\frac{M_1}{M_1 \cap I^{n+1}M_2} \to \frac{M_1}{M_1 \cap I^n M_2}$$

are surjective.

It remains to identify

$$\varprojlim \frac{M_1}{M_1 \cap I^n M_2}$$

with $\widehat{M_1}$.

Since A is Noetherian and $M_1 \subseteq M_2$ with M_2 finitely generated, the Artin–Rees lemma implies that there exists c such that for all $n \geq c$,

$$M_1 \cap I^n M_2 = I^{n-c}(M_1 \cap I^c M_2).$$

In particular, for $n \geq c$,

$$I^n M_1 \subseteq M_1 \cap I^n M_2 \subseteq I^{n-c} M_1.$$

Thus the filtrations

$$\{I^n M_1\} \qquad \text{and} \qquad \{M_1 \cap I^n M_2\}$$

are equivalent, hence define the same completion. Therefore

$$\varprojlim \frac{M_1}{M_1 \cap I^n M_2} \cong \varprojlim \frac{M_1}{I^n M_1} = \widehat{M_1}.$$

So we obtain the exact sequence

$$0 \to \widehat{M_1} \to \widehat{M_2} \to \widehat{M_3} \to 0.$$

(c) The canonical map $M \to \widehat{M}$ and its kernel.

Define

$$\iota_M : M \to \widehat{M} = \varprojlim_n M/I^n M$$

by

$$\iota_M(m) = \big(m \bmod I^n M\big)_{n \geq 0}.$$

This is clearly an A-module homomorphism.

An element $m \in M$ lies in the kernel if and only if its image in every quotient $M/I^n M$ is zero, i.e. if and only if

$$m \in I^n M \qquad \text{for every } n \geq 0.$$

Hence

$$\ker(\iota_M) = \bigcap_{n=0}^{\infty} I^n M.$$

(d) The map $\widehat{A} \otimes_A M \to \widehat{M}$.

Since $\widehat{M}$ is naturally an $\widehat{A}$-module, the rule

$$(\hat{a}, m) \longmapsto \hat{a} \cdot \iota_M(m)$$

defines an A-balanced map

$$\widehat{A} \times M \to \widehat{M}.$$

Hence there is an induced homomorphism

$$\theta_M : \widehat{A} \otimes_A M \to \widehat{M},$$

which is the composition stated in the problem.

Assume now that M is finitely generated. Let

$$\pi : A^r \to M$$

be a surjection. Take any element

$$x = (x_n) \in \widehat{M} = \varprojlim_n M/I^n M.$$

For each n, the induced map

$$(A/I^n)^r \to M/I^n M$$

is surjective, so choose $a^{(n)} \in (A/I^n)^r$ mapping to x_n. Because the transition maps

$$(A/I^{n+1})^r \to (A/I^n)^r$$

are surjective, the lifts may be chosen compatibly. Thus they determine an element of

$$\varprojlim_n (A/I^n)^r = \widehat{A}^r.$$

Its image in $\widehat{M}$ is x. Therefore the natural map

$$\widehat{A}^r \to \widehat{M}$$

is surjective. Since this map factors through

$$\widehat{A} \otimes_A M,$$

it follows that

$$\theta_M : \widehat{A} \otimes_A M \to \widehat{M}$$

is surjective.

Assume moreover that A is Noetherian. Choose a finite presentation

$$A^m \xrightarrow{u} A^r \to M \to 0.$$

Tensoring with $\widehat{A}$ gives a right exact sequence

$$\widehat{A}^m \to \widehat{A}^r \to \widehat{A} \otimes_A M \to 0.$$

Hence $\widehat{A} \otimes_A M$ is the cokernel of

$$\widehat{A}^m \to \widehat{A}^r.$$

Let $K = \mathrm{im}(u) \subseteq A^r$. Since A is Noetherian, K is finitely generated. Applying part (b) to

$$0 \to K \to A^r \to M \to 0$$

gives
$$0 \to \widehat{K} \to \widehat{A}^r \to \widehat{M} \to 0.$$
Applying part (b) again to
$$0 \to \ker(u) \to A^m \to K \to 0$$
shows that
$$\widehat{A}^m \to \widehat{K} \to 0$$
is exact, so the image of
$$\widehat{A}^m \to \widehat{A}^r$$
is exactly $\widehat{K}$. Therefore $\widehat{M}$ is also the cokernel of
$$\widehat{A}^m \to \widehat{A}^r.$$

Thus both $\widehat{A} \otimes_A M$ and $\widehat{M}$ are the cokernel of the same map, and so
$$\widehat{A} \otimes_A M \xrightarrow{\sim} \widehat{M}.$$

Therefore, if M is finitely generated, θ_M is surjective, and if in addition A is Noetherian, θ_M is an isomorphism.

Deep theory pack for Problems 6–9: inverse limits, filtrations, and I-adic completion

1. Inverse systems and inverse limits

Definition. An inverse system of abelian groups is a diagram
$$G_1 \xleftarrow{\phi_1} G_2 \xleftarrow{\phi_2} G_3 \xleftarrow{\phi_3} \cdots$$
consisting of groups G_n and homomorphisms
$$\phi_n : G_{n+1} \to G_n.$$

The inverse limit is the object that collects all compatible elements in this tower.

Concrete construction. The inverse limit is
$$\varprojlim G_n = \left\{ (x_1, x_2, \dots) \in \prod_{n \geq 1} G_n \;\middle|\; \phi_n(x_{n+1}) = x_n \text{ for all } n \right\}.$$
It is a subgroup of the direct product, with componentwise operations.

If the G_n are rings, then $\varprojlim G_n$ is a subring of $\prod G_n$, with multiplication
$$(x_n)(y_n) = (x_n y_n).$$
If the G_n are A-modules, then $\varprojlim G_n$ is an A-submodule of $\prod G_n$.

Universal property. The inverse limit $G = \varprojlim G_n$ comes equipped with projection maps
$$g_n : G \to G_n$$

such that

$$\phi_n \circ g_{n+1} = g_n.$$

Moreover, if H is another abelian group with maps

$$h_n : H \to G_n$$

satisfying

$$\phi_n \circ h_{n+1} = h_n$$

for all n, then there is a unique homomorphism

$$h : H \to G$$

such that

$$h_n = g_n \circ h$$

for all n.

This universal property characterizes $\varprojlim G_n$ uniquely up to unique isomorphism.

2. Exactness properties of inverse limits

Suppose we have inverse systems $\{A_n\}$, $\{B_n\}$, $\{C_n\}$ and short exact sequences

$$0 \to A_n \to B_n \to C_n \to 0$$

for each n, compatible with the transition maps.

Then one always gets an exact sequence

$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n.$$

Thus inverse limit is *left exact*.

However, surjectivity on the right may fail in general. The issue is that compatible elements in $\varprojlim C_n$ may admit lifts in each B_n, but these lifts need not be compatible across n.

A useful sufficient condition for surjectivity is that the transition maps on the left,

$$A_{n+1} \to A_n,$$

are all surjective. In that case one can adjust lifts inductively and obtain exactness

$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0.$$

More generally, the relevant abstract condition is the Mittag–Leffler condition, of which surjective transition maps are a special case.

3. Filtrations and the induced topology

Let G be an abelian group with a descending filtration

$$G \supset G_1 \supset G_2 \supset \cdots.$$

Then there are natural quotient maps

$$G/G_{n+1} \to G/G_n,$$

so we obtain an inverse system

$$G/G_1 \leftarrow G/G_2 \leftarrow G/G_3 \leftarrow \cdots .$$

This filtration defines a topology on G: a basis of open sets is given by cosets

$$g + G_n.$$

The neighborhoods of 0 are precisely the subgroups G_n. This is a typical linear topology.

Hausdorff condition. This topology is Hausdorff if and only if

$$\bigcap_{n\geq 1} G_n = 0.$$

Indeed, in a topological group, the closure of $\{0\}$ is the intersection of all neighborhoods of 0.

Cauchy sequences. A sequence (g_m) is Cauchy if for every n there exists N such that

$$g_m - g_{m'} \in G_n \qquad (m, m' \geq N).$$

Thus the images of g_m in G/G_n eventually stabilize.

Two Cauchy sequences (g_m) and (g'_m) are equivalent if for every n, eventually

$$g_m - g'_m \in G_n.$$

The completion $\widehat{G}$ is defined as the set of Cauchy sequences modulo this equivalence.

Key identification. There is a canonical isomorphism

$$\widehat{G} \cong \varprojlim G/G_n.$$

Indeed, a Cauchy sequence determines for each n an eventual class in G/G_n, and these classes are compatible. Conversely, a compatible family of classes admits representatives whose sequence is Cauchy.

This shows the completion depends only on the induced topology, not on the particular choice of filtration giving that topology.

4. I-adic topology and completion

Let A be a ring and $I \subseteq A$ an ideal. Then

$$A \supset I \supset I^2 \supset I^3 \supset \cdots$$

is a filtration, giving the inverse system

$$A/I \leftarrow A/I^2 \leftarrow A/I^3 \leftarrow \cdots .$$

Its inverse limit is

$$\widehat{A} = \varprojlim A/I^n,$$

called the I-adic completion of A.

Similarly, for an A-module M,

$$M \supset IM \supset I^2M \supset I^3M \supset \cdots$$

gives

$$\widehat{M} = \varprojlim M/I^nM,$$

the I-adic completion of M.

There are canonical maps

$$A \to \widehat{A}, \qquad M \to \widehat{M},$$

sending an element to its compatible system of residue classes modulo I^n.

For modules,

$$m \mapsto (m \bmod I^nM)_n.$$

Its kernel is

$$\ker(M \to \widehat{M}) = \bigcap_{n\geq 0} I^nM.$$

Thus M is called I-adically separated if

$$\bigcap_{n\geq 0} I^nM = 0,$$

and I-adically complete if

$$M \to \widehat{M}$$

is an isomorphism.

5. Fundamental examples

Formal power series. Let

$$A = k[x_1, \dots, x_n], \qquad I = (x_1, \dots, x_n).$$

Then

$$\widehat{A} = \varprojlim A/I^r \cong k[[x_1, \dots, x_n]].$$

Indeed, modulo I^r one remembers only monomials of total degree $< r$, so the inverse limit is precisely the ring of formal power series

$$\sum_{\alpha\in\mathbf{N}^n} a_\alpha x^\alpha.$$

p-adic integers. Let

$$A = \mathbf{Z}, \qquad I = (p).$$

Then

$$\widehat{A} = \varprojlim \mathbf{Z}/p^n\mathbf{Z} = \mathbf{Z}_p.$$

An element of $\mathbf{Z}_p$ is a compatible family of residues modulo p^n, equivalently a formal expansion

$$a_0 + a_1p + a_2p^2 + \cdots, \qquad a_i \in \{0, \dots, p-1\}.$$

Thus $\mathbf{Z}_p$ is the ring of p-adic integers. Its field of fractions is $\mathbf{Q}_p$, the field of p-adic numbers.

6. Exactness of completion and the Artin–Rees lemma

Inverse limits are only left exact in general, so completion is not automatically exact in arbitrary situations. The Noetherian finite-generation hypothesis is what restores good exactness properties.

Artin–Rees lemma. Let A be Noetherian, M a finitely generated A-module, and $N \subseteq M$ a submodule. Then there exists $c \geq 0$ such that for all sufficiently large n,

$$N \cap I^n M = I^{n-c}(N \cap I^c M).$$

In particular, the filtration $\{N \cap I^n M\}$ is equivalent to the intrinsic filtration $\{I^n N\}$.

This lemma is the key tool for comparing completions of submodules inside larger modules.

Exactness theorem. If A is Noetherian and

$$0 \to M_1 \to M_2 \to M_3 \to 0$$

is an exact sequence of finitely generated A-modules, then

$$0 \to \widehat{M_1} \to \widehat{M_2} \to \widehat{M_3} \to 0$$

is exact.

Thus completion preserves short exact sequences in the Noetherian finitely generated setting.

7. Tensor description of completion

For any A-module M, there is a natural homomorphism

$$\widehat{A} \otimes_A M \to \widehat{M}.$$

It comes from the natural action of $\widehat{A}$ on $\widehat{M}$.

If M is finitely generated, then this map is surjective.

If moreover A is Noetherian, then it is an isomorphism:

$$\widehat{A} \otimes_A M \xrightarrow{\sim} \widehat{M}.$$

So for finitely generated modules over a Noetherian ring, completion is exactly extension of scalars from A to $\widehat{A}$.

This is closely related to the flatness of $\widehat{A}$ over A in the Noetherian setting.

8. Separatedness, Krull intersection, and geometric meaning

A natural question is when

$$\bigcap_{n \geq 0} I^n M = 0.$$

In many Noetherian local situations this is ensured by the Krull intersection theorem, especially when I lies in the Jacobson radical. Then the canonical map

$$M \to \widehat{M}$$

is injective.

Geometrically, I-adic completion describes infinitesimal neighborhoods. For example, if $I = (x_1, \dots, x_n)$, the completion records all formal Taylor expansions at the origin:

$$k[x_1, \dots, x_n]^\wedge \cong k[[x_1, \dots, x_n]].$$

Thus completion algebraically formalizes the idea of zooming infinitely close to a point or subvariety.

9. Global picture of Problems 6–9

Problems 6–9 form one coherent theory.

Problem 6: construct inverse limits concretely and prove they exist.
Problem 7: prove inverse limit is left exact, and becomes exact on the right under surjectivity hypotheses.
Problem 8: interpret filtrations topologically and identify completion with

$$\varprojlim G/G_n.$$

Problem 9: specialize to the I-adic setting, obtaining formal power series rings, p-adic integers, exactness of completion, and the tensor formula

$$\widehat{A} \otimes_A M \cong \widehat{M}$$

for finitely generated modules over Noetherian rings.

So the overall chain of ideas is

$$\text{inverse systems} \Longrightarrow \text{inverse limits} \Longrightarrow \text{filtration topologies} \Longrightarrow \text{completions} \Longrightarrow I\text{-adic algebra}.$$

Concrete worked examples pack for Problems 6–9

1. Example: $\mathbf{Z}_p = \varprojlim \mathbf{Z}/p^n\mathbf{Z}$

Let

$$A = \mathbf{Z}, \qquad I = (p).$$

Then the I-adic completion is

$$\widehat{A} = \varprojlim_n \mathbf{Z}/p^n\mathbf{Z} = \mathbf{Z}_p.$$

An element of $\mathbf{Z}_p$ is a compatible family

$$(a_1, a_2, a_3, \dots), \qquad a_n \in \mathbf{Z}/p^n\mathbf{Z},$$

such that

$$a_{n+1} \equiv a_n \pmod{p^n}.$$

Example. For $p = 5$, the compatible family

$$2 \bmod 5, \qquad 7 \bmod 25, \qquad 57 \bmod 125, \qquad 182 \bmod 625, \dots$$

defines an element of $\mathbf{Z}_5$, since

$$7 \equiv 2 \pmod 5, \qquad 57 \equiv 7 \pmod{25}, \qquad 182 \equiv 57 \pmod{125}.$$

The canonical map

$$\mathbf{Z} \to \mathbf{Z}_p, \qquad n \mapsto (n \bmod p^r)_{r \geq 1}$$

has kernel

$$\bigcap_{r \geq 1} p^r \mathbf{Z} = 0.$$

Thus $\mathbf{Z} \hookrightarrow \mathbf{Z}_p$ is injective.

2. Example: $k[[x]] = \varprojlim k[x]/(x^n)$

Let

$$A = k[x], \qquad I = (x).$$

Then

$$\widehat{A} = \varprojlim_n k[x]/(x^n) = k[[x]].$$

An element of $k[x]/(x^n)$ is represented by a truncated polynomial

$$a_0 + a_1 x + \cdots + a_{n-1} x^{n-1}.$$

A compatible family of truncations is exactly a formal power series

$$a_0 + a_1 x + a_2 x^2 + \cdots .$$

Example. The compatible family

$$1 + x \pmod{x^2}, \qquad 1 + x + x^2 \pmod{x^3}, \qquad 1 + x + x^2 + x^3 \pmod{x^4}, \qquad \ldots$$

defines the element

$$1 + x + x^2 + x^3 + \cdots \in k[[x]].$$

Multiplication is formal: if

$$f(x) = \sum_{i \geq 0} a_i x^i, \qquad g(x) = \sum_{j \geq 0} b_j x^j,$$

then

$$f(x)g(x) = \sum_{n \geq 0} \left(\sum_{i+j=n} a_i b_j \right) x^n.$$

3. Example: $k[[x, y]] = \varprojlim k[x, y]/(x, y)^n$

Let

$$A = k[x, y], \qquad \mathfrak{m} = (x, y).$$

Then

$$\widehat{A} = \varprojlim_n k[x, y]/\mathfrak{m}^n = k[[x, y]].$$

Modulo $\mathfrak{m}^n$, only monomials of total degree $< n$ survive. For example,

$$k[x,y]/(x,y)^3$$

has representatives of the form

$$a + bx + cy + dx^2 + exy + fy^2.$$

Thus an element of $k[[x,y]]$ is a compatible family of finite truncations, i.e. a formal power series

$$\sum_{i,j\geq 0} a_{ij}x^iy^j.$$

Geometrically, this is the formal local ring at the origin.

4. Example: the canonical map $M \to \widehat{M}$

Let

$$A = k[x], \qquad I = (x), \qquad M = A.$$

Then

$$\widehat{M} = k[[x]].$$

The canonical map is

$$k[x] \to k[[x]],$$

sending a polynomial to the same expression viewed as a formal series.

Its kernel is

$$\bigcap_{n\geq 0} x^n k[x] = 0,$$

so the map is injective.

This completion adds genuinely new elements, for example

$$1 + x + x^2 + x^3 + \cdots \in k[[x]]$$

does not lie in $k[x]$.

5. Example: module completion of $M = A/(x^2, y)$

Let

$$A = k[x,y], \qquad I = (x,y), \qquad M = A/(x^2,y).$$

In M, we have

$$y = 0, \qquad x^2 = 0,$$

so every element is of the form

$$a + bx.$$

Hence

$$M \cong k \oplus kx$$

as a k-vector space.

Now

$$IM = (x,y)M = xM = kx,$$

and
$$I^2M = 0.$$
Therefore for every $n \geq 2$,
$$M/I^nM = M.$$
So the inverse system stabilizes, and
$$\widehat{M} \cong M.$$

This illustrates that a module may already be complete if its I-adic filtration stabilizes.

6. Example: completion of a quotient

Let
$$A = k[x], \qquad I = (x), \qquad M = A/(x^2).$$
Then
$$\widehat{A} = k[[x]].$$
Since A is Noetherian and M is finitely generated,
$$\widehat{M} \cong \widehat{A} \otimes_A M \cong k[[x]] \otimes_{k[x]} k[x]/(x^2) \cong k[[x]]/(x^2).$$

Thus $\widehat{M}$ consists of elements
$$a + bx \qquad (a, b \in k),$$
because $x^2 = 0$.

Another example is
$$A = k[x, y], \qquad I = (x, y), \qquad M = A/(y - x^2),$$
for which
$$\widehat{M} \cong k[[x, y]]/(y - x^2).$$

7. Example of exactness after completion

Let
$$A = k[x], \qquad I = (x),$$
and consider the exact sequence
$$0 \to (x) \to A \to A/(x) \to 0.$$
Completing x-adically gives
$$0 \to (x) \to k[[x]] \to k \to 0,$$
which is exact.

This is the simplest example of exactness of completion in the Noetherian finitely generated setting.

8. Example of $\widehat{A} \otimes_A M \cong \widehat{M}$

Let
$$A = k[x], \qquad I = (x), \qquad M = A/(x^2).$$
Then
$$\widehat{A} = k[[x]]$$
and
$$\widehat{A} \otimes_A M = k[[x]] \otimes_{k[x]} k[x]/(x^2) \cong k[[x]]/(x^2).$$
On the other hand,
$$\widehat{M} \cong k[[x]]/(x^2).$$
Hence explicitly
$$\widehat{A} \otimes_A M \cong \widehat{M}.$$

9. Separated and non-separated examples

Separated example. For
$$A = k[x], \qquad I = (x), \qquad M = A,$$
we have
$$\bigcap_{n \geq 0} x^n A = 0,$$
so $A \to \widehat{A}$ is injective.

Non-separated toy example. If a filtered abelian group G has
$$G_n = G \qquad \text{for all } n,$$
then
$$\bigcap_n G_n = G,$$
so the topology is not Hausdorff, and separatedness fails completely.

10. Final conceptual summary

These examples illustrate the general theory:

- $\mathbf{Z}_p$ is obtained from compatible residues modulo p^n,
- $k[[x]]$ is obtained from compatible polynomial truncations modulo x^n,
- $k[[x, y]]$ is obtained from compatible truncations modulo powers of (x, y),
- module completions may stabilize if nilpotence appears,
- exactness is preserved after completion in the Noetherian finitely generated setting,
- and
$$\widehat{A} \otimes_A M \cong \widehat{M}$$
shows that completion of finitely generated modules is extension of scalars to the completed ring.

So completion means passing from finite algebraic approximations to one infinite formal object encoding all of them at once.

Comparison and diagram pack for Problems 6–9

1. Big roadmap diagram

The general flow of ideas is

inverse system $\Longrightarrow$ inverse limit $\Longrightarrow$ filtration topology $\Longrightarrow$ completion $\Longrightarrow$ I-adic completion.

More concretely, an inverse system

$$G_1 \leftarrow G_2 \leftarrow G_3 \leftarrow \cdots$$

gives

$$\varprojlim G_n.$$

If the system comes from a filtration

$$G \supset G_1 \supset G_2 \supset \cdots,$$

then

$$G/G_1 \leftarrow G/G_2 \leftarrow G/G_3 \leftarrow \cdots$$

and

$$\widehat{G} \cong \varprojlim G/G_n.$$

If the filtration is I-adic on a ring A, then

$$A/I \leftarrow A/I^2 \leftarrow A/I^3 \leftarrow \cdots$$

and

$$\widehat{A} = \varprojlim A/I^n.$$

Similarly for an A-module M:

$$M/IM \leftarrow M/I^2M \leftarrow M/I^3M \leftarrow \cdots$$

and

$$\widehat{M} = \varprojlim M/I^nM.$$

2. Diagram: inverse limit as compatible sequences

One may picture the inverse limit as a subset of the product:

$$\varprojlim G_n \subseteq \prod_{n\geq 1} G_n.$$

Its elements are sequences

$$(x_1, x_2, x_3, \dots)$$

such that

$$\phi_1(x_2) = x_1, \qquad \phi_2(x_3) = x_2, \qquad \phi_3(x_4) = x_3, \ \dots$$

Thus:

$$x_1 \quad \leftarrow \quad x_2 \quad \leftarrow \quad x_3 \quad \leftarrow \cdots$$

$$G_1 \quad \xleftarrow{\phi_1} \quad G_2 \quad \xleftarrow{\phi_2} \quad G_3 \quad \xleftarrow{\phi_3} \cdots$$

3. Diagram: completion from a filtration

Let
$$G \supset G_1 \supset G_2 \supset \cdots .$$
Then
$$G/G_1 \leftarrow G/G_2 \leftarrow G/G_3 \leftarrow \cdots$$
is an inverse system.

A Cauchy sequence (g_m) determines eventual classes modulo each G_n:
$$(g_m) \rightsquigarrow (\text{eventual class mod } G_1, \text{eventual class mod } G_2, \dots),$$
hence an element of
$$\varprojlim G/G_n.$$

Conversely, a compatible system of classes gives a Cauchy sequence of lifts. Therefore
$$\widehat{G} \cong \varprojlim G/G_n.$$

4. Canonical maps

There are natural maps
$$G \to \widehat{G}, \qquad A \to \widehat{A}, \qquad M \to \widehat{M}.$$

For a filtered group G,
$$g \longmapsto [(g, g, g, \dots)], \qquad \ker(G \to \widehat{G}) = \bigcap_{n \geq 1} G_n.$$

For a ring A with ideal I,
$$a \longmapsto (a \bmod I^n)_n, \qquad \ker(A \to \widehat{A}) = \bigcap_{n \geq 0} I^n.$$

For an A-module M,
$$m \longmapsto (m \bmod I^n M)_n, \qquad \ker(M \to \widehat{M}) = \bigcap_{n \geq 0} I^n M.$$

5. Exactness diagram

Suppose
$$0 \to A_n \to B_n \to C_n \to 0$$
is exact for every n, compatibly with inverse-system maps. Then
$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n$$
is always exact.

So inverse limit is left exact.

If the transition maps $A_{n+1} \to A_n$ are surjective, then
$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0$$
is exact.

6. Comparison table: $\varprojlim$, $\widehat{G}$, $\widehat{A}$, $\widehat{M}$

Object	Starting data	Definition	Meaning
$\varprojlim G_n$	inverse system $G_{n+1} \to G_n$	compatible sequences (x_n)	all compatible approximati
$\widehat{G}$	$G \supset G_1 \supset G_2 \supset \cdots$	Cauchy sequences mod equivalence	completion of filtered
$\widehat{A}$	ring $A,\ I \subseteq A$	$\varprojlim A/I^n$	I-adic completion c
$\widehat{M}$	A-module $M,\ I \subseteq A$	$\varprojlim M/I^nM$	I-adic completion o

7. Comparison table: topology viewpoint

Object	Neighborhoods of 0	Hausdorff / separated	Completion
G	G_n	$\bigcap_n G_n = 0$	$\widehat{G}$
A	I^n	$\bigcap_n I^n = 0$	$\widehat{A}$
M	I^nM	$\bigcap_n I^nM = 0$	$\widehat{M}$

8. Comparison table: main examples

Input	Filtration	Completion
$\mathbf{Z},\ I = (p)$	$p^n\mathbf{Z}$	$\mathbf{Z}_p$
$k[x],\ I = (x)$	(x^n)	$k[[x]]$
$k[x_1, \ldots, x_n],\ I = (x_1, \ldots, x_n)$	I^r	$k[[x_1, \ldots, x_n]]$
$A/(x^2)$ over $A = k[x]$	$x^n(A/(x^2))$	$k[[x]]/(x^2)$
$A/(x^2, y)$ over $A = k[x, y]$	$(x, y)^nM$	$\widehat{M} \cong M$

9. Comparison table: $\mathbf{Z}_p$ versus $k[[x]]$

Feature	$\mathbf{Z}_p$	$k[[x]]$
base ring	$\mathbf{Z}$	$k[x]$
ideal	(p)	(x)
quotients	$\mathbf{Z}/p^n\mathbf{Z}$	$k[x]/(x^n)$
completion	compatible residues mod p^n	compatible truncations mod x^n
formal expansion	$a_0 + a_1p + a_2p^2 + \cdots$	$a_0 + a_1x + a_2x^2 + \cdots$
topology	p-adic	x-adic

Thus p in $\mathbf{Z}_p$ plays the same role as x in $k[[x]]$.

10. Comparison table: separated versus complete

Property	Meaning
separated	$M \to \widehat{M}$ is injective
complete	$M \to \widehat{M}$ is an isomorphism
separated condition	$\bigcap_{n \geq 0} I^nM = 0$
complete condition	every compatible system in $\varprojlim M/I^nM$ comes from M

11. Comparison table: exactness and tensor description

Statement	When true
$0 \to \widehat{M_1} \to \widehat{M_2} \to \widehat{M_3}$ exact	from left exactness in inverse-limit setting
surjectivity on the right	needs extra hypotheses
$0 \to \widehat{M_1} \to \widehat{M_2} \to \widehat{M_3} \to 0$	A Noetherian, M_i finitely generated
$\widehat{A} \otimes_A M \to \widehat{M}$ surjective	M finitely generated
$\widehat{A} \otimes_A M \cong \widehat{M}$	A Noetherian, M finitely generated

12. Formal neighborhood picture

For

$$A = k[x], \qquad I = (x),$$

the quotients

$$A/(x),\ A/(x^2),\ A/(x^3), \ldots$$

record higher and higher infinitesimal information near the point $x = 0$.

For

$$A = k[x, y], \qquad \mathfrak{m} = (x, y),$$

the quotients

$$A/\mathfrak{m},\ A/\mathfrak{m}^2,\ A/\mathfrak{m}^3, \ldots$$

record more and more information near the origin $(0, 0)$.

Their inverse limit is

$$k[[x, y]],$$

the formal local picture at the origin.

So one may summarize:

$$\text{functions near a point} \Longrightarrow \text{jets of all orders} \Longrightarrow \text{formal power series.}$$

13. Final summary chart

$$\boxed{\varprojlim(\text{successive quotients}) = \text{completion}}$$

and in the I-adic case:

$$\boxed{\widehat{A} = \varprojlim A/I^n, \qquad \widehat{M} = \varprojlim M/I^n M.}$$

Main examples:

$$\boxed{\mathbf{Z}_p = \varprojlim \mathbf{Z}/p^n\mathbf{Z}, \qquad k[[x_1, \ldots, x_n]] = \varprojlim k[x_1, \ldots, x_n]/(x_1, \ldots, x_n)^r.}$$